

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Total Kjeldahl Nitrogen	WQP 013	1.6	

PREPARED BY			
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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC
1.1	7/21/23	Adjusted buffer concentrations following conversation with SEAL	TAC
1.2	9/8/23	Increased sample volume and changed to acid addition in the lab according to QAPP.	TAC
1.3	11/30/23	Instructions to include the digestion of standards. Corrected TKN calibrant concentrations. Addition of sodium chloride to Sampler Wash Solution. General SOP revision for process clarification or to highlight important steps.	TAC
1.4	6/10/24	Corrected TKN calibrant concentrations. Calibrants, CRM, and Spike concentration (mg/L) changed. The digestion section was changed to include the digestion of calibrants, CRM, blanks, and spikes. The instrument preparation section was moved before running the instrument start-up. Addition of instructions to retrieve report as ASCII file.	TAC
1.4.1	7/22/24	Corrected accuracy to match correct value in QAPP	TAC
1.5	12/03/24	Reagent Revision - Copper sulfate pentahydrate was replaced with Copper II sulfate. Added details for the digestion Process (Calibrant and Spikes	TAC

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		amounts) and the dissolution of tube contents process. Eliminated Spike preparation instructions for 5 and 10 mL volumes. Added details throughout the SOP to improve readability.	
1.6	6/6/2024	Minor edits to improve clarity.	TAC

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1. Title

Standard Operating Procedure for Total Kjeldahl Nitrogen (Kjeldahl Digestion) in Seawater.

2. Applicability

This SOP applies to the determination of seawater Total Kjeldahl Nitrogen (TKN) at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater bottom TKN at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater TKN data following the Quality Assurance Project Plan.

4. Method

- 4.1. EPA Method 351.2 (Environmental Protection Agency)

5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. Read each reagent SDS prior to use for safe handling and disposal.
- 5.3. Prepare reagents to be used on this SOP inside a hood.
- 5.4. Wear disposable latex gloves, lab coat, and safety glasses while performing this procedure.
- 5.5. Be aware of waste container liquid levels to avoid spills.
- 5.6. **Be extremely careful while handling the block digester to prevent any accident with boiling acid or an accidental burn by the digester block.**

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.
- 6.3. Strictly follow SOPs and manufacturer instructions.
- 6.4. Follow weekly cleaning instructions closely to avoid precipitation inside the reagent lines.
- 6.5. Ensure the supply of reagents and wash is adequate to complete the run.
- 6.6. **Avoid reagent precipitation in the lines by closely following instructions related to the salicylate line tube insertion/removal from reagents bottles.**
- 6.7. Never attempt to turn off or change pump speed during a run, unless there is an emergency.
- 6.8. Never leave the pump platen engaged without pumping for long periods of time.
- 6.9. The photometer LED lamp must be changed with the instrument turned OFF.
- 6.10. Keep all instrument covers in place during analysis.

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6.11. Ensure the waste tube is never submerged in the waste.

7. Training

- 7.1. Personnel shall be trained in the proper sample collection, handling, and analysis procedure for TKN.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Field Sampling Log Form (Appendix F)
- 8.2. Consumables
 - 8.2.1. Kimwipes
 - 8.2.2. Latex Gloves
 - 8.2.3. Beaker (1000 mL) for solution preparation
 - 8.2.4. Glass bottles for reagent and wash storage (1000mL)
 - 8.2.5. Plastic bottles for reagent and wash storage (100 mL)
 - 8.2.6. Amber plastic/glass bottle (1000 mL)
 - 8.2.7. Falcon tube (50 mL) for sample collection
 - 8.2.8. Plastic syringe 25 mL
 - 8.2.9. PES Filters (0.45 µM)
- 8.3. Digestion Unit
 - 8.3.1. Digestion Tubes
 - 8.3.2. Kjeldahl Digestion Granules
 - 8.3.3. Tear Drop Stoppers
- 8.4. Vortex Mixer
- 8.5. AA500 Autoanalyzer
- 8.6. Required Chemicals

Compound	CAS No.	Safety classification	
Ammonium Sulfate (NH ₄) ₂ SO ₄	7783-20-2	GHS07	Irritant
Dichloroisocyanuric acid sodium salt, C ₃ Cl ₂ N ₃ NaO ₃ · 2H ₂ O	51580-86-0	GHS03, GHS07, GHS09	Danger - Oxidizing, Irritant, Warning/Danger- Environmentally Damaging
Ethylenediamine tetra-acetic acid disodium salt, C ₁₀ H ₁₄ N ₂ Na ₂ O ₈ · 2H ₂ O	6381-92-6	GHS07	Irritant, Warning/Danger-
Iso-propanol, C ₃ H ₇ OH	67-63-0	GHS02, GHS07	Danger – Flammable, Irritant, Warning/Danger-
Sodium chloride, NaCl	7647-14-5	--	
Sodium hydroxide, NaOH	1310-73-2	GHS05	Warning/Danger

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Compound (Cont.)	CAS No.	Safety classification	
Sodium hydrogen phosphate-12-hydrate, $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$	10039-32-4	--	
Sodium hypochlorite, 10% solution, NaClO	7681-52-9	GHS05, GHS09	Warning/Danger – Corrosive, Environmentally Damaging
Sodium nitroprusside, $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] \cdot 2\text{H}_2\text{O}$	13755-38-9	GHS06	Danger - Toxic
Sodium salicylate, $\text{C}_7\text{H}_5\text{NaO}_3$	54-21-7	GHS07	Irritant
Sulfuric acid, conc., H_2SO_4	7664-93-9	GHS05	Warning/Danger - Corrosive
Potassium sulfate, K_2SO_4	7778-80-5	–	
Copper (II) sulfate, CuSO_4	7758-98-7	GHS05, GHS07, GHS09	Warning/Danger – Corrosive, Irritant, Warning/Danger - Environmentally Damaging

8.7. Reagents, Standards, and Preparation Instructions

8.7.1. Sampler Wash Solution

07-12: Sample Wash Solution		
Reagent	Amount	Instructions
Sodium chloride (NaCl)	35 g	1. Weigh the Copper sulfate and the Sodium chloride. 2. Place a beaker on a stir plate and add the weighed compounds. 3. Dissolve the beaker contents in about 800 mL of Distilled Water. 4. Measure the Sulfuric acid using a graduated cylinder (50 mL) and a pipette (3,600 uL) 5. Carefully add the Sulfuric acid to the beaker. 6. Transfer the solution to a graduated cylinder and bring it to volume with Distilled water.
Sulfuric acid, conc. (H ₂ SO ₄)	53.6 mL	
Copper (II) sulfate (CuSO ₄) (7758-98-7)	2.92 g	
Distilled water	1000 mL	
General Details		
Storage	Glass or plastic container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	Stock standard solutions preparation and sampler solution.	

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8.7.2. Synthetic Seawater

Reagent	Amount	Instructions	
Sodium chloride (NaCl)	35 g	1. Weigh sodium chloride. 2. Add the measured amount into a beaker with a stir rod. 3. Add approximately 500-600 mL of distilled water to the beaker. 4. Stir on a plate until all the salt has dissolved. 5. Pour beaker contents into a 1L graduated cylinder. 6. Bring to volume with distilled water.	
Distilled Water	1L		
General Details			
Storage	Plastic container at room temperature.		
Stability	One week.		
Purpose	Stock standard solutions preparation.		

8.7.3. Digestion Solution

0.7.5: Digestion solution		
Reagent	Amount	Instructions
Sulfuric acid (H ₂ SO ₄)	134 mL	1. Weigh the ingredients. 2. Transfer the weighed materials to a beaker. 3. Add approximately 500 mL of distilled water to the beaker. 4. Place the beaker on a stir plate and mix. 5. Carefully add the sulfuric acid. 6. Continue stirring until completely dissolved. 7. Transfer to a graduated cylinder. 8. Bring to volume (1 L) with distilled water.
Potassium sulfate (K ₂ SO ₄) (7778-80-5)	134 g	
Copper II Sulfate (CuSO ₄) (7758-98-7)	7.3 g	
Distilled Water	1000 mL	
General Details		
Storage	Glass container at room temperature.	
Stability	Indefinite.	
Purpose	Digestion solution to be added to the sample during the digestion process.	

8.7.4. Special Wash Solution

Reagent	Amount	Instructions
Sodium 10% hypochlorite	10 mL	1. Add 10 mL of Sodium hypochlorite to a graduated cylinder. 2. Bring to volume with distilled water.
Distilled water	50 mL	
General Details		
Storage	Plastic container at room temperature.	
Stability	One day.	
Purpose	For weekly cleanup.	

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8.7.5. Hydrochloric acid (1 mol/L)

Reagent	Amount	Instructions	
Hydrochloric acid (36.5-38%)	85 mL	1. Add approximately 700 mL of Distilled Water into a graduated flask. 2. Measure HCl and carefully add it to the graduated cylinder. 3. Let it cool to room temperature. 4. Dilute to volume and mix. 5. Store in a plastic bottle.	
Distilled water	1000 mL		
General Details			
Storage	Plastic container at room temperature.		
Stability	Indefinite. Replace if turbid.		
Purpose	For weekly cleanup.		

8.7.6. Buffer Solution

Reagent	Amount	Instructions
EDTA (6381-92-6)	9.20 g	1. Weigh all the ingredients. 2. Add all the ingredients to a beaker. 3. Add approximately 200 mL of Distilled Water. 4. Place the beaker on a stir plate and mix. 5. Transfer the volume to a graduated cylinder. 6. Bring to volume (500 mL) with Distilled Water.
Sodium hydroxide	11 g	
Sodium hydrogen phosphate-12-hydrate (10039-32-4)	20 g	
Distilled water	500 mL	
General Details		
Storage	Plastic container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	For TKN analysis.	

8.7.7. Sodium Salicylate

Reagent	Amount	Instructions
Sodium Salicylate (54-21-7)	158 g	1. Weigh sodium nitroprusside, add it to a beaker with approximately 200 mL of distilled water, and stir. (It takes a while to dissolve.) 2. Add the weighed sodium salicylate to the beaker. 3. Transfer the volume to a graduated cylinder. 4. Bring to volume (500 mL) with distilled water.
Sodium nitroprusside (13755-38-9)	0.4 g	
Distilled water	500mL	
General Details		
Storage	Amber plastic container at room temperature.	
Stability	One week.	
Purpose	For TKN analysis.	

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8.7.8. Dichloroisocyanurate

Reagent	Amount	Instructions
Dichloroisocyanuric acid sodium salt dihydrate (51580-86-0)	2.25 g	1. Add approximately 200 mL of distilled water to the beaker. 2. Weigh the Dichloroisocyanuric salt. 3. Add the salt to the beaker. 4. Place the beaker on a stir plate and mix. 5. Bring to volume (500 mL) with distilled water.
Distilled water	500 mL	
General Details		
Storage	Amber glass container at room temperature.	
Stability	One week.	
Purpose	For TKN analysis.	

8.7.9. Stock Standard A, 100 mmol/L

Reagent	Amount	Instructions
Ammonium sulfate (7783-20-2)	0.660 g	1. Weigh the Ammonium sulfate. 2. Add approximately 50 mL of distilled water to a beaker. 3. Add the weighted ingredient to the beaker. 4. Place the beaker on a stir plate and mix. 5. Bring to volume (100 mL) with synthetic seawater.
Synthetic seawater	100 mL	
General Details		
Storage	Glass or plastic container at 4°C.	
Stability	One year.	
Purpose	Stock standard for calibration.	

8.7.10. Stock Standard B, 10 mmol/L

Reagent	Amount	Instructions
Stock standard A	10 mL	1. Pipette stock standard A. 2. Pour the measured amount into a 100 mL volumetric flask. 3. Bring to volume with synthetic seawater. 4. Mix well.
Synthetic seawater	100 mL	
General Details		
Storage	Glass or plastic container at 4°C.	
Stability	One year.	
Purpose	Stock standard for calibration.	

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8.7.11. Working standards

8.7.11.1. Prepare working standards fresh before each analysis run.

8.7.11.2. Pipette **Stock Standard B** into 100 mL volumetric flasks, following the table below.

8.7.11.3. Dilute to volume with **Synthetic Seawater** and mix thoroughly.

mL of Stock Standard B (Low Range)	µL of Stock Standard B (Low Range)	Calibrants [mg/L]
1.6	1600	2.22
1.2	1200	1.66
0.8	800	1.11
0.4	400	0.55
0.16	160	0.22

9. Procedure

9.1. Sample Collection

9.1.1. Prepare a Van Dorn bottle to be deployed.

9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.

9.1.3. Drop the bottle messenger to close the Van Dorn bottle at the desired depth.

9.1.4. Pull the Van Dorn bottle back up into the boat.

9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.

9.1.6. Push open the Van Dorn bottle's spigot valve to allow water flow.

9.1.7. Dispense a few mL of the seawater into a clean 50 mL sample tube to rinse the tube 3 times.

9.1.8. Fill the 50 mL sample tube with seawater.

9.1.9. Place the sample bottle inside the cooler until analysis in the lab.

9.1.9.1. Samples can be preserved for 28 days at 4°C prior to analysis.

9.1.9.2. **Poison the sample** upon arrival in the lab.

9.1.9.2.1. Carefully add **200 µL** of 5.4 N sulfuric acid (H_2SO_4) to the 50 mL sample tube.

9.1.9.2.2. Close the Falcon tube and invert to mix.

9.1.9.3. Analyze samples within 28 days.

9.2. Kjeldahl Sample Digestion

Never run digestion alone. At least two people shall be present during the digestion process to ensure all steps are followed correctly and safe operation of the block digester.

9.2.1. Prepare the digester rack tubes.

9.2.1.1. Preparation of tubes on the rack shall be done prior to inserting the rack into the digester block.

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- 9.2.1.2. Label the digestion tubes with a permanent marker.
- 9.2.1.3. Load the rack with labeled tubes.
- 9.2.2. Pipette **25 mL of samples/calibrants/blank/CRM/spikes** and transfer the measured volumes to the corresponding digestion tubes.
 - 9.2.2.1. The blank is the synthetic seawater (**prepare 2 tubes**)
 - 9.2.2.2. Prepare calibrants following the instructions in Section 8.7.11.
 - 9.2.2.2.1. Prepare **2 tubes** with the highest calibrant.
 - 9.2.2.3. Prepare CRM following the instructions in Section 13.3.
 - 9.2.2.3.1. Prepare **2 tubes** with the CRM.
 - 9.2.2.4. Prepare Spikes following the instructions in Section 13.4
 - 9.2.2.4.1. Spikes are prepared for the blank and samples (refer to Tray protocol setup section 9.3.2.5.3)
 - 9.2.2.5. Record the digestion in the “TKN Digestion Log” for future reference.
- 9.2.3. Add **10 mL of digestion solution** to each tube.
- 9.2.4. Mix with a vortex mixer.
- 9.2.5. **Add five Digestion Granules** to each tube.
 - 9.2.5.1. Do not exceed this amount, as this may cause the sample to boil over.
- 9.2.6. **STOP!** Ensure all prior steps were completed before proceeding and that every tube has digestion granules in it.
- 9.2.7. Place the digester rack in the digester block.
- 9.2.8. Turn ON the digester controller.
 - 9.2.8.1. Ensure the digester block is connected to the controller; otherwise, the controller will not turn on.
 - 9.2.8.2. The digester cable should be connected to a 240 VAC outlet (located on the opposite side of the wall of the fume hood)
- 9.2.9. Press “Run” to select the Controller Program to be used for the day’s run.
 - 9.2.9.1. Make sure the controller displays “Program 1”; if not, scroll through the programs using the side arrows.
 - 9.2.9.2. Program 1 contains the following steps:
 - 9.2.9.2.1. The block digester will preheat to 160°C.
 - 9.2.9.2.2. The digester will pause at 1 hour and produce a beeping sound for the teardrop stoppers to be inserted in the tube's mouth.
 - 9.2.9.2.3. After pressing “Cont”, the digester will continue to heat until 380°C for 1.5 hours.
 - 9.2.9.2.4. After the digestion time is completed, the block digester powers off.
- 9.2.10. Press “Run” to start Program 1.
- 9.2.11. Wait 1 hour for a beeping sound and carefully add a teardrop stopper to each tube.
- 9.2.12. Press “Cont”.
 - 9.2.12.1. The digester will proceed to 380°C for 1.5 hours.
 - 9.2.12.2. The end of the digestion process will produce a beeping sound, the message “Program Complete” will be displayed on the controller screen, and the off

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icon (square with a dot in the center) will be visible at the upper right corner of the display window.

- 9.2.13. Press the “OK” key to deactivate the alarm.
- 9.2.14. Let the block cool for at least several hours until the temperature is close to room temperature, ~25-30°C, before proceeding.
- 9.2.15. Carefully handle tubes and add **25 mL of distilled water** to each tube.
- 9.2.16. Re-insert tubes into the digestion block to warm water and help loosen the tube contents. This may take several hours.
- 9.2.17. Press “Run” twice to start “Program 2”.
 - 9.2.17.1. If “program 2” is not displayed in the controller window, search for the program using the side arrows.
- 9.2.18. Remove one tube at a time and mix the tube contents using a vortex mixer.
 - 9.2.18.1. Continue this step until all granules have loosened from the bottom of the tube.
- 9.2.19. Transfer digested volumes into 50 mL tubes and label accordingly.
- 9.2.20. **Filter the sample** using a 0.45 µm PES filter directly into a 15mLFalcon tube.
- 9.2.21. Proceed to prepare the instrument for analysis following instructions in Section 9.3.
- 9.2.22. Stopping a Program, if necessary.
 - 9.2.22.1. Press the “Stop” button at any time to turn off the block.
- 9.2.23. Pausing a program, if necessary.
 - 9.2.23.1. Press “Pause” button at any time to pause the program.
 - 9.2.23.2. Press “Cont” button to continue running the program.
- 9.2.24. Unplug the block digester when the system is not in use.

9.3. Sample/Instrument Pre-Run Preparation: Analysis Setup

- 9.3.1. Prepare Sampler tubes according to the Analysis Protocol with calibration standards and previously filtered samples.
 - 9.3.1.1. Place the highest calibration standard (50 mL tube) in the calibrant rack on the left-hand side (Numbers 921 or 922).
 - 9.3.1.2. Pour Calibration Standards into the sampler cups (or 15 mL falcon tubes) and position them into the calibrant rack on the left-hand side (Numbers 901-920).
 - 9.3.1.3. Pour each sample to be analyzed into the sampler cups (or 15 mL falcon tubes) and position into the sample racks on the right-hand side (Numbers 1-120).
- 9.3.2. Create a New Run or Copy Run
 - 9.3.2.1. Click “Set Up”
 - 9.3.2.2. Click “Analysis/Run”
 - 9.3.2.3. Double-click on the “TKN” folder from the left side of the pop-up window.
 - 9.3.2.4. Select the TKN.anl run from the list in the center of the window.
 - 9.3.2.5. Click “New Run” or “Copy Run” to create a run based on the previously selected template.

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- 9.3.2.5.1. Insert Run Name: “Date_Quarter#_TKNRun#” in the provided space at the upper right-hand corner of the Main tab. (E.g. 231025_Q2_TKN1)
- 9.3.2.5.2. Check Channel tab
 - 9.3.2.5.2.1. Calibrant concentrations and units (mg/L)
 - 9.3.2.5.2.2. Spike Recovery Standard Concentration [139 mg/L].
- 9.3.2.5.3. Modify the Tray Protocol tab to match your run according to the following criteria:
 - 9.3.2.5.3.1. 1 Primer (P) and 1 Drift (D) with the highest calibrant.
 - 9.3.2.5.3.2. Calibrants + Blank (C) in decreasing order.
 - 9.3.2.5.3.3. Baseline (B), whose cup number is 0.
 - 9.3.2.5.3.4. 3 Carryovers; 1 H1 using the highest calibrant and 2 L1 using the lowest calibrant.
 - 9.3.2.5.3.5. 1 Recovery - Spiked Blanks
 - 9.3.2.5.3.6. 1 CRM as QC1
 - 9.3.2.5.3.7. 10 Samples
 - 9.3.2.5.3.8. 1 Spiked Sample (K).
 - 9.3.2.5.3.9. 1 CRM as QC1
 - 9.3.2.5.3.10. Baseline (B), whose cup number is 0
 - 9.3.2.5.3.11. Repeat 10 samples, 1 QC, and 1 Baseline until completing the sample analysis for the day.
 - 9.3.2.5.3.12. Ensure that the tray protocol ends with 1 Drift (D) using the highest calibrant, and 1 End (E), which is the cup number 0.
- 9.3.2.5.4. Create a New Run based on a previously existing Tray Protocol:
 - 9.3.2.5.4.1. Import a previously created tray protocol from Excel. Make sure the tray protocol follows the correct format to be imported.
 - 9.3.2.5.4.2. Excel files must be closed before importing.
- 9.3.2.5.5. Click “OK” to save a new run.

9.4. Autoanalyzer Instrument Start-Up

Note: Avoid reagent precipitation in the lines by closely following instructions related to the salicylate line insertion/removal from the liquid stream.

- 9.4.1. Check LED and change if necessary.
 - 9.4.1.1. Note: The instrument needs to be **OFF** to perform this step.
 - 9.4.1.2. Remove the photometer cover of Channel 1.
 - 9.4.1.3. Ensure the lamp wavelength corresponds to TKN analysis.
 - 9.4.1.3.1. TKN lamp: **660 nm**.
 - 9.4.1.3.2. If not, proceed to change the lamp following the instructions below.
 - 9.4.1.4. Unscrew the light power cable.

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- 9.4.1.5. Loosen the two screws that hold the LED.
- 9.4.1.6. Replace the LED in place with the 660nm lamp.
- 9.4.1.7. Ensure the lamp is correctly placed over the holder and fasten the screws.
- 9.4.1.8. Carefully insert and tighten the light power cable.
- 9.4.1.9. Close the photometer cover.
- 9.4.2. Ensure an adequate supply of reagents by checking the reagent container levels.
- 9.4.3. Ensure the Channel 1 lines are correctly connected to the pump module.
Lines are labeled close to the pump module, and numbers may be found engraved on a silver plate on the right side of the pump module, prior to entering the system.
 - 9.4.3.1. **Line A** to pump line #10
 - 9.4.3.2. **Line B** to pump line #6
 - 9.4.3.3. **Line C** to pump line #8
- 9.4.4. Ensure the waste disposal line from Channel 1 is inside a waste tube by itself.
- 9.4.5. Turn ON the instrument in the following order.
 - 9.4.5.1. Computer
 - 9.4.5.2. Sampler
 - 9.4.5.3. Power Supply
- 9.4.6. Open the AACE Software.
- 9.4.7. Login - No password
 - 9.4.7.1. Click "OK".
- 9.4.8. Confirm the analyzer and software are communicating by observing a permanent blue light in the photometer module, meaning it is ready for operation.
 - 9.4.8.1. Blinking blue light means the system is operating (charting & running).
 - 9.4.8.2. Red - STOP! Leak detected.
- 9.4.9. Place the reagent lines inside the distilled water bottle
 - 9.4.9.1. Blue line labeled "TKN Dichloro..."
 - 9.4.9.2. Clear line labeled "TKN Buffer"
 - 9.4.9.3. Wash line - clear thick line labeled "Channel 1 Wash"
 - 9.4.9.4. Red line labeled "TKN Salicylate"
 - 9.4.9.5. Clear line - labeled "distilled water"
 - 9.4.9.6. All Channel 3 lines should also be placed inside the distilled water bottle
- 9.4.10. Place the sampler wash/sample line into the distilled water bottle.
 - 9.4.10.1. This line is the only one to the right side of the AS2 sampler (close to the computer station).
- 9.4.11. Click "Charting" on the system window toolbar.
 - 9.4.11.1.1. Or click on "Run" in the main menu and select "Chart".
 - 9.4.11.2. Click "OK" if the TKN analysis is displayed or "Change" to select the TKN analysis.
 - 9.4.11.2.1. Double-click on "TKN".
 - 9.4.11.2.2. Double-click on the "TKN.anl" file from the panel to the center of the window.
- 9.4.12. Click on the "Reagent Valve Set 1" icon on the system's main window.

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- 9.4.12.1. Select “TKN_no salicylate.rva”.
- 9.4.12.2. Click “OK”.
- 9.4.13. Click on the “Pump” icon on the system’s main window.
 - 9.4.13.1. Select “Normal” to pump at normal speed.
 - 9.4.13.2. Select “Engage Platen”.
 - 9.4.13.2.1. Observe the platen moving down slowly. When in place, the LED light on the control switch will turn green.
 - 9.4.13.3. Press “Close” to close the window.
 - 9.4.13.3.1. The pump will continue running at the selected speed.
- 9.4.14. Set Light Power.

Perform this adjustment every time an analysis is run for the first time, when pump tubes get changed, and occasionally to compensate for lamp aging.

 - 9.4.14.1. Right-click on the charting window.
 - 9.4.14.2. Select “Set Light Power” from the pop-up menu.
- 9.4.15. Ensure the Gain is set to 10.
 - 9.4.15.1. Gain value can be seen at the bottom of the charting window.
 - 9.4.15.2. If not set to 10, change the value by clicking on “Channel 1” on the system’s main window and manually inserting the right value in the provided space for the Gain.
- 9.4.16. Perform Water Baseline Check.
 - 9.4.16.1. Pump System Wash Solution/distilled water through the system for 10-15 minutes at normal speed.
 - 9.4.16.1.1. Visually inspect the bubble pattern through the whole system for the same size, shape, and distance. See AA500 Complete page 6.6 for bubble visual reference.
 - 9.4.16.1.2. Monitor flowchart and wait for the system to stabilize.
 - 9.4.16.1.2.1. If the baseline is not visible, wait a few minutes and continue with the next step.
 - 9.4.16.2. **Set the base** once the system is stable.
 - 9.4.16.2.1. Right-click on the chart.
 - 9.4.16.2.2. Select “Set Base/Offset”.
 - 9.4.16.2.3. Refer to “Wash Solution Baseline” in the “How To” Section if baseline does not stabilize.
 - 9.4.16.3. Annotate the water baseline value (%) on the chart window’s upper right corner.
 - 9.4.16.3.1. Expected value is 5.0 %
- 9.4.17. Perform Reagent Check.
 - 9.4.17.1. Remove the sampler wash/sample line from the distilled water bottle and pat dry the line with a Kimwipe.
 - 9.4.17.1.1. Insert the sampler wash/sample line in the Sampler Wash Solution bottle.

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- 9.4.17.2. Remove reagent lines from the distilled water bottle, pat dry the tubes with a Kimwipe, and insert each line into its corresponding reagent solution container.
 - 9.4.17.2.1. Red line labeled “TKN Salicylate” into the Sodium Salicylate bottle
 - 9.4.17.2.2. Blue line labeled “TKN Dichloro...” into the Dichloroisocyanurate bottle
 - 9.4.17.2.3. Clear line labeled “TKN Buffer” into the Buffer bottle
 - 9.4.17.2.4. Leave the Channel 1 wash line inside the distilled water bottle.
 - 9.4.17.2.5. Follow system diagrams for reference.
- 9.4.17.3. Cover all reagent bottles with parafilm.
- 9.4.17.4. Continue pumping solutions through reagent lines at normal speed for **5 minutes**.
 - 9.4.17.4.1. Visually inspect the bubble pattern and allow it to stabilize.
- 9.4.17.5. **Connect Salicylate line.**
 - 9.4.17.5.1. Click on the “Reagent Valve Set 1” icon on the system’s main window.
 - 9.4.17.5.2. Select “TKN.rva”.
 - 9.4.17.5.3. Click “OK”.
- 9.4.17.6. Visually inspect the bubble pattern and allow it to stabilize.
 - 9.4.17.6.1. Yellowish/Greenish color will start to form after the salicylate line entrance.
- 9.4.17.7. Wait for the baseline to stabilize and annotate the new baseline %.
 - 9.4.17.7.1. After observing the salicylate baseline
- 9.4.17.8. Calculate the reagent absorbance.
 - 9.4.17.8.1. Insert the reagent baseline value (%) and the gain (10) into the corresponding space in the “Absorbance_Calcs.” worksheet located in the following path: Desktop>AA500 Docs&Manuals>DailyUse.
 - 9.4.17.8.2. If the resulting absorbance is >20%, reagents, sampler wash solution, or wash pot may be contaminated.
- 9.4.17.9. **Reset the Base.**
 - 9.4.17.9.1. Right-click on the chart screen.
 - 9.4.17.9.2. Select “Set Base/Offset”.
- 9.4.17.10. Wait several minutes to ensure the baseline is stable.
- 9.4.18. Perform Sensitivity (Gain) Check.
 - 9.4.18.1. Sample the highest calibrant
 - 9.4.18.1.1. Double-click on the “Sampler” icon.
 - 9.4.18.1.2. Enter the cup number containing the highest calibrant.
 - 9.4.18.1.3. Click on “Sample”.
 - 9.4.18.1.4. Pump for 2 minutes.
 - 9.4.18.1.5. After 2 minutes, select “Wash”.

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- 9.4.18.1.6. Close the window by clicking “Cancel”.
- 9.4.18.2. **Observe the Primer peak (1st peak) until it reaches a plateau.**
 - 9.4.18.2.1. This may take ~8 minutes.
 - 9.4.18.2.2. Annotate the peak value %.
- 9.4.18.3. Right-click on the chart screen and select “**Set Gain**” while the peak is still at the plateau.
- 9.4.18.4. Calculate the Sensitivity
 - 9.4.18.4.1. Complete the Sensitivity table in the “Absorbance_Calcs” worksheet with the corresponding values (peak %, gain, baseline %, highest calibrant).
 - 9.4.18.4.2. The peak % used is the one observed after setting the gain (~90%)
- 9.4.18.5. **Wait until the peak fully passes and reset the base.**
 - 9.4.18.5.1. Right-click on the chart screen and select “**Set Base/Offset**”.
- 9.4.18.6. Verify the baseline and bubble pattern stability.
- 9.4.19. Proceed to analyze samples.

9.5. Start the Run

- 9.5.1. **Verify that all sample IDs refer to the correct sample rack position in the Tray Protocol before proceeding.**
- 9.5.2. Click on “Run” in the main menu.
- 9.5.3. Click on “Start”.
- 9.5.4. Double-click on the “Data” folder under the “Analysis” area.
- 9.5.5. Double-click on “TKN”.
- 9.5.6. Select the previously created .run file from the list in the center of the window.
- 9.5.7. Click “OK”.
- 9.5.8. Insert the analyst's name and comments in the provided spaces on the pop-up window.
- 9.5.9. Click “OK” to start analyzing samples.
 - 9.5.9.1. Sample protocol will start running and will stop once the last command on the Tray Protocol has been run.
 - 9.5.9.2. Wait for a pop-up window with a run completion message on the top left corner of the computer window.
- 9.5.10. Continue with Daily Shutdown instructions in Section 9.6.

9.6. Daily Shutdown

Perform this Daily shutdown if the instrument will be used for a few consecutive days otherwise, complete a Weekly Cleanup (Refer to Section 11 for instructions.)

- 9.6.1. Restart Charting.
 - 9.6.1.1. Once the system ends a run, it closes the charting windows and disables some software functions.
- 9.6.2. **Remove the Salicylate line** from the reagent bottle and pat it dry.

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- 9.6.3. Insert the Salicylate line tube into the distilled water bottle and **pump at fast mode for 5 minutes.**
- 9.6.4. **After 5 minutes, remove the Sampler Wash line** from the Sampler Wash bottle, pat it dry, and insert the Sampler Wash line into the distilled water bottle. Wait a couple of minutes.
- 9.6.5. Switch lines:
 - 9.6.5.1. Remove reagent lines from reagent bottles and pat them dry with a Kimwipe. **(Remove the Buffer solution line last)**
 - 9.6.5.2. Insert reagent lines into the distilled water bottle.
- 9.6.6. Pump for a minimum of 20 minutes.
- 9.6.7. After 20 minutes, remove all reagent lines/sampler lines from the distilled water bottle.
- 9.6.8. Pat dry the tubes and store them inside the provided flasks.
- 9.6.9. Cover the sampler wash pot with a Kimwipe or parafilm.
- 9.6.10. Drain the system by continuing to pump. Observe the lines in the chemistry module until most bubbles have exited the system.
- 9.6.11. Turn off the pump and release the platen.
 - 9.6.11.1. Click on the “Pump” icon on the main window.
 - 9.6.11.2. Switch the pump off.
 - 9.6.11.3. Click on “Release platen”.
- 9.6.12. Stop Charting.
 - 9.6.12.1. Click the “Stop” button in the System window toolbar.
 - 9.6.12.1.1. Or click the “Close” button in the charting window.
- 9.6.13. Turn OFF the instrument in the following order.
 - 9.6.13.1. Power Supply
 - 9.6.13.2. Sampler
 - 9.6.13.3. Computer
- 9.6.14. Discard samples and standards from calibrant/sample cups in the provided waste container.
- 9.6.15. Wash sample cups.
 - 9.6.15.1. Rinse cups with tap water 3 times.
 - 9.6.15.2. Submerge the cups in a 10% HCl bath and let soak.
 - 9.6.15.3. After use, properly store any remaining HCl to be reused several times.
 - 9.6.15.3.1. Discard into the corresponding unwanted material container if turbid.
 - 9.6.15.4. Triple-rinse with Distilled Water.
 - 9.6.15.5. Let the cups dry standing upside down on a dish rack or inside the oven.

9.7. Data Retrieval

- 9.7.1. Reports
 - 9.7.1.1. Click on “Retrieve” on the main menu.
 - 9.7.1.2. Select “Report” from the dropdown.

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- 9.7.1.3. Double-click on the desired analysis (menu on the left window).
- 9.7.1.4. Double-click on the desired .run File.
- 9.7.1.5. Select the desired option from the buttons at the bottom of the report window.
 - 9.7.1.5.1. You may save a PDF version of the report at this point.
- 9.7.2. ASCII files
 - 9.7.2.1. Click on “Retrieve” on the main menu.
 - 9.7.2.2. Select “View Chart” from the dropdown.
 - 9.7.2.3. Double-click on the desired analysis (menu on the left window).
 - 9.7.2.4. Double-click on the desired .run File.
 - 9.7.2.5. On the icons on top of the chart window, click on the “Export to ASCII file” icon (3rd left to right)
 - 9.7.2.6. Format Tab
 - 9.7.2.6.1. Format should always be set to “Text Format (ASCII)”
 - 9.7.2.6.2. Confirm the Directory to be used
 - 9.7.2.7. Export Data Tab
 - 9.7.2.7.1. Check all desired information to be included in the file
 - 9.7.2.7.2. Go to the “Preview Tab” to confirm the selection
 - 9.7.2.8. Click on “Export” to save the file.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Calibration data will be recorded in the Autoanalyzer station lab notebook.
- 10.3. Field forms and Maintenance sheets shall be kept in the designated file cabinet at the CCRI office.
- 10.4. Daily/Weekly/Monthly/Yearly maintenance shall be recorded in maintenance sheets.
- 10.5. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.6. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.7. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. The autoanalyzer exterior may be cleaned using an alcohol-damped lint-free cloth.
- 11.2. Refer to AA500 Complete User Manual for specific instrument maintenance instructions.
- 11.3. Refer to BD50/BD28 Block Digestion System User Manual for maintenance instructions for the digester.
- 11.4. Recommended pump tubes use is for a maximum of 200 hours.
- 11.5. Autoanalyzer Daily Cleanup
 - 11.5.1. Follow the instructions provided under Section 9.6 Daily Shutdown.
 - 11.5.2. Document your cleanup method in the provided lab notebook.
- 11.6. Autoanalyzer Weekly Cleanup

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- 11.6.1. Select “All Wash” Mode by pressing the reagent valve set 1 icon
- 11.6.2. Pump the following through all reagent lines (“All Wash Mode” at fast mode for at least 2 mins each.
 - 11.6.2.1. NaOH (Sodium hydroxide)
 - 11.6.2.2. Distilled water
 - 11.6.2.3. HCl 1N (Hydrochloric acid)
 - 11.6.2.4. Distilled water
 - 11.6.2.5. Special Wash Solution 10% (Sodium hypochlorite)
 - 11.6.2.6. Distilled water
- 11.6.3. Remove the lines from the distilled water bottle and store them dry inside their respective channel storage flask.
- 11.6.4. Leave the pump running for a few minutes to drain the system.
 - 11.6.4.1. Observe until no liquid is seen inside the system and/or the waste line.
- 11.6.5. Once the system has drained, stop Charting.
 - 11.6.5.1. Click the “Stop” button in the System window toolbar.
 - 11.6.5.1.1. Or click the “Close” button in the charting window.
- 11.6.6. Print and fill out the Weekly Maintenance Sheet included in the AA500 Complete User Manual.
- 11.6.7. Save the sheet in the instrument’s maintenance binder.
- 11.7. For Monthly and Yearly Maintenance, follow the instructions provided in the User Manual and complete the corresponding documentation.
- 11.8. Digester Maintenance
 - 11.8.1. Periodically clean the digester covers, the block, and the controller keypad with diluted mild detergent and a soft cloth.
 - 11.8.2. Clean any spills right away.

12. Troubleshooting

- 12.1. Refer to “AA500 Complete User Manual” (Section 8) For Troubleshooting instructions.
- 12.2. Refer to AA500 Complete User Manual (Section 6.10) for sensitivity and noise troubleshooting instructions.
- 12.3. For problems with the digester, refer to Section 9 of the “BD50/BD28 Block Digestion System User Manual”.

13. QA/QC

- 13.1. As per QAPP Table 4, seawater TKN should be measured in mg/L with an accuracy of 0.5 mg/L ($\pm 20\%$) and a precision of $<20\%$.
- 13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.
- 13.3. CRM Dilution
 - 13.3.1. Pipette 150 μ L (0.15 mL) of TKN CRM [1000 mg/L] solution and bring to volume with synthetic seawater in a 100 mL volumetric flask. This dilution will have a concentration of **1.5 mg/L**.

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13.4. **Spike** Preparation for Digestion (total vol. = 25 mL)

13.4.1. Pipette 24,820 µL of Synthetic Sea Water/Sample into a digestion tube

13.4.1.1. 2x 10mL

13.4.1.2. 1x 4,000 µL

13.4.1.3. 1x 820 µL

13.4.2. Add 180 µL of Stock B [139 mg/L] into a digestion tube

13.4.3. This will add **1.0 mg/L** TKN to the sample.

14. References

- 14.1. U.S. ENVIRONMENTAL PROTECTION AGENCY. 1993. Methods for Determination of Inorganic Substances in Environmental Samples; EPA-600/R/93/100—Draft. Environmental Monitoring Systems Lab., Cincinnati, Ohio.
- 14.2. AA500 Complete User Manual.
- 14.3. AA500 Operational Manual.
- 14.4. AACE Software Manual.
- 14.5. BD50/BD28 Block Digestion System User Manual.